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Response of secondary production by macroinvertebrates to large wood addition in three Michigan streams

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SUMMARY

1. We measured responses in macroinvertebrate secondary production after large wood additions to three forested headwater streams in the Upper Peninsula of Michigan. These streams had fine-grained sediments and low retention capacity due to low amounts of in-channel wood from a legacy of past logging. We predicted that wood addition would increase macroinvertebrate secondary production by increasing exposed coarse substrate and retention of organic matter.
2. Large wood (25 logs) was added haphazardly to a 100-m reach in each stream, and a 100-m upstream reach served as control; each reach was sampled monthly, 1 year before and 2 years after wood addition (i.e. BACI design). Macroinvertebrate secondary production was measured 1 year after wood addition in two habitat types: inorganic sediments of the main channel and debris accumulations of leaf litter and small wood.
3. Overall macroinvertebrate production did not change significantly because each stream responded differently to wood addition. Production increased by 22% in the main-channel of one stream, and showed insignificant changes in the other two streams compared to values before wood addition. Changes in main-channel macroinvertebrate production were related to small changes in substrate composition, which probably affected habitat and periphyton abundance. Macroinvertebrate production was much greater in debris accumulations than in the main-channel, indicating the potential for increased retention of leaf litter to increase overall macroinvertebrate production, especially in autumn.
4. Surrounding land use, substrate composition, temperature and method of log placement are variables that interact to influence the response of stream biota to wood additions. In most studies, wood additions occur in altered catchments, are rarely monitored, and secondary production is not a common metric. Our results suggest that the time required for measurable changes in geomorphology, organic matter retention, or invertebrate production is likely to take years to achieve, so monitoring should span more than 5 years, and ecosystem metrics, such as macroinvertebrate secondary production, should be incorporated into restoration monitoring programs.

Keywords: headwater streams, large wood, macroinvertebrates, restoration, secondary production

Introduction

The primary goal of ecosystem restoration is to restore as much taxonomic and functional diversity as practical while also enhancing ecosystem function (Alexander & Allan, 2006). Many obstacles to this goal arise as managers try to meet specific restoration objectives, including identifying and measuring the most relevant ecological metrics at the appropriate frequency and ecological scale. Functional processes occur at varying rates at different scales, and are

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constrained by abiotic and biotic ''filters'' that interact locally, as well as climatic and geological 'filters' that interact at larger spatial and temporal scales (Poff & Ward, 1990; Levin, 1992). Therefore, ecosystem restorations should include monitoring of clearly defined functional variables measured at spatial and temporal scales appropriate for the stated management goals (Moerke *et al.*, 2004; Palmer *et al.*, 2005).

At a global scale, small streams have been widely degraded by changes in land use and associated impacts, with concurrent and alarming losses in freshwater biodiversity and impaired ecological function (Lake, 2005; Meyer *et al.*, 2007; Nilsson *et al.*, 2007). Streams that drain logged, urbanised or agricultural watersheds often exhibit habitat simplification, reduced water quality, loss of species and increased susceptibility to invasive species (Lodge, 1993; Meyer & Wallace, 2001; Abell, Allan & Lehner, 2007). Headwater streams are commonly subject to wood removal and channelisation to increase land area for human use, often resulting in increased flooding and habitat destruction (Kail, 2002; Woolsey *et al.*, 2007). The biological effects of such habitat simplification include declines in fish, amphibian and invertebrate diversity (Triska, 1984; Pretty *et al.*, 2003).

Countries in the European Union (EU), Australia and the U.S.A. are spending the equivalent of billions of U.S. dollars annually to increase riparian buffers, improve water quality, restore natural hydrologic regimes and enhance habitat in streams and rivers (e.g. Gerhard & Reich, 2000; Bernhardt *et al.*, 2005; Lake, 2005). Yet, shockingly few stream and river restorations are ever monitored beyond visual evaluation. Although the number of monitored restorations increased from about 1% in the 1980s to 6% in the 1990s, over 90% of restorations are still not monitored and most restoration projects are not designed as experiments with testable hypotheses (Bernhardt *et al.*, 2005; Kail *et al.*, 2007).

Forest practices such as timber harvest and wood removal have reduced the quantity and quality of stream habitat at a global scale (Triska, 1984; Kail, 2002; Gurnell *et al.*, 2005). Restoration involving the addition of large wood has been shown to reduce sedimentation in some streams (Waters, 1995) and increase invertebrate diversity, biomass and trophic complexity in others (Wallace, Webster & Meyer, 1995; Lemly & Hilderbrand, 2000). However, little is known about how large wood influences stream ecosystem function, and macroinvertebrate secondary production in particular. In some streams, large wood has been shown to increase the retention of terrestrially derived (allochthonous) organic matter, thereby making it available to stream biota (Johnson, Breneman & Richards, 2003; Lepori, Palm & Malmqvist, 2005). The addition of large wood can also alter stream geomorphology through substrate sorting, which may expose gravel in the main channel and deposit sand along the banks of the stream channel (Kail, 2002). Changes in local geomorphology could alter organic matter processing and macroinvertebrate secondary production by increasing stable habitat and possibly changing food availability for biota (Smock, Metzler & Gladden, 1989; Lemly & Hilderbrand, 2000; Benke & Wallace, 2003).

Only one previous study in a single stream has examined how wood addition changed food resources and macroinvertebrate community structure and production (Wallace *et al.*, 1995), even though wood additions have been identified as a global priority for the improvement of stream habitat. Measurements of secondary production incorporate changes in density, growth and biomass at the population level over time (Benke & Huryn, 2006) and overcome limitations of static community metrics. Secondary production is especially useful in the context of restoration, because it estimates the total amount of heterotrophic biomass produced over a given period, including energy transferred to higher trophic levels (Waters, 1979; Huryn & Wallace, 2000).

Our objective was to assess changes in macroinvertebrate secondary production following an experimental large wood addition to three headwater streams in managed forests of the Upper Peninsula of Michigan, U.S.A. We measured macroinvertebrate secondary production in an upstream control reach and downstream treatment reach for 1 year before and 2 years after experimental wood addition to the three streams. We quantified secondary production in the main channel for the entire study period, and in debris accumulations seasonally for 1 year following wood addition. We predicted that the addition of large wood would: (i) alter local geomorphology and thereby change the functional composition of secondary production in the main channel, and (ii) increase organic matter retention and thus indirectly increase secondary production of detritivores.

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Methods

Study area

We studied three first-order streams (Shane Creek, State Creek, and Walton Creek) located in the Ottawa National Forest in the Ontonagon River basin of the Upper Peninsula of Michigan (Fig. 1; 89º1′W, 46°28′N). In each stream we designated a 100-m upstream control reach and a 100-m treatment reach, separated by a 30–50 m buffer reach. All three streams drain small watersheds of second-growth forest with relatively intact riparian vegetation. The Shane Creek watershed is 83% forested and was last logged in 1967, State Creek is 95% forested and was also last logged in 1967, and Walton Creek is 85% forested and was last logged in 1915. The remainder of each watershed is primarily wetland. Riparian trees include hemlock (*Tsuga canadenensis* L.), white pine (*Pinus alba* L.), red maple (*Acer rubrum* L.), sugar maple (*Acer saccharum* Marsh.), and paper birch (*Betula papyfera* L.), which are typical riparian species in the region (Smith 1995). State Creek had a thick understory of tag alder (*Alnus serrulata* Alt.). Prior to manipulation, all three streams had low amounts of large wood (Table 1), low standing stocks of coarse benthic organic matter (CBOM; Cordova *et al.*, 2007; Entrekin *et al.*, 2007; Hoellein *et al.*, 2007), and litter fall rates comparable to streams in deciduous forests (Entrekin *et al.*, 2008). After 1 year of pre-treatment data collection, in May 2004 we haphazardly added 25 large big tooth aspen logs (*Populus grandidentata*, each 2.5 m long × 0.5 m diameter) to the 100-m treatment reach in each of the three streams, and continued to collect data for 2 years post-treatment. Control and treatment reaches were separated by 30–50 m.

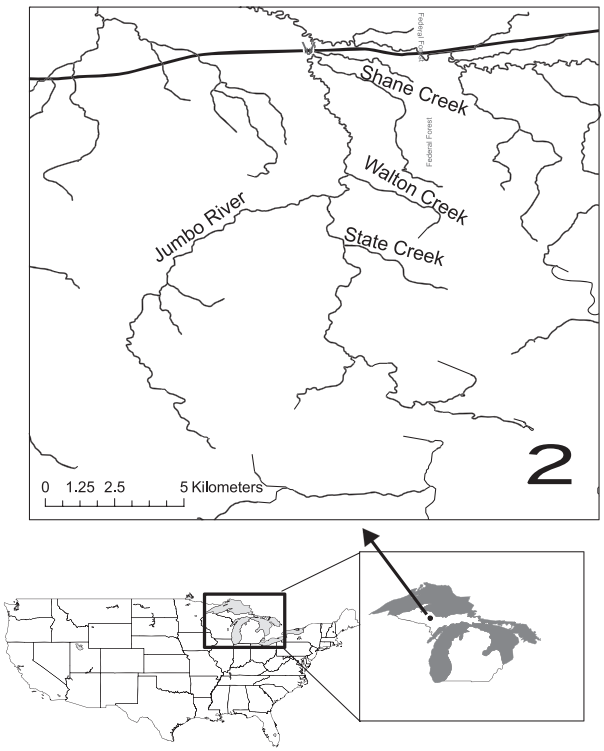

Fig. 1 Study area of experimental streams in the Upper Peninsula of Michigan, U.S.A. from Hoellein *et al.*, 2007.

Organic matter – main channel

We quantified CBOM and fine benthic organic matter (FBOM) standing stocks in five 804-cm^2 cores taken from each stream reach. We first removed CBOM, including leaf litter, using a 1-mm sieve, and then subsampled the FBOM slurry by mixing sediment in the corer and recording total and subsample volume. After transport to the laboratory on ice, CBOM was oven-dried at 60 °C, sorted by organic matter type (leaves, moss and wood), and weighed. A subsample of each organic matter type was combusted at 550 °C and reweighed to determine ash-free dry mass (AFDM; Benfield, 2006). For FBOM, we filtered the slurry subsample onto a glass fiber filter, which was dried at 60 °C, weighed, combusted at 550 °C, and reweighed to determine AFDM.

Organic matter – debris accumulations

Debris accumulations represented a low proportion of total habitat; however, we realised their importance to invertebrates after we began pretreatment sampling. Because most benthic sampling devices cannot effectively sample debris accumulations, we sampled debris accumulations using a novel method and a seasonal sampling regime that began 1 year after wood addition (June and November 2004; April, July, and November 2005; April 2006). First, we quantified the volume of all debris accumulations in each reach by measuring length and width parallel and perpendicular to the stream channel and depth. We also noted if the debris accumulation was associated with our added logs (i.e. ''new'' logs). Within an

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Table 1 Average (±SE) annual physical and chemical attributes of the study reaches. Y0 = year before wood addition (May 2003–April 2004) and Y1-Y2 are 2 years after wood addition (May 2004–May 2006). All substrata are not presented, and therefore percentages will not sum to 100

	Shane Creek				State Creek				Walton Creek			
	Y0		Y1-Y2		Y0		Y1-Y2		Y0		Y1-Y2	
	C	T	C	T	C	T	C	T	C	T	C	T
Temperature (°C)	7.2		6.3		6.6		6.4		6.39		7.9	
Litter fall (g AFDM m^{-1} day^{-1})	8.7	11.8	–	–	9.8	5.9	–	–	4.8	6.6	–	–
Slope (%)	1.0		0.8		2.4		1.2		0.8		1.2	
Bankful width (m)	2.3	2.4	2.5	2.5	2.1	2.3	2.2	1.8	1.9	1.8	2.2	2.4
Large wood (m^3 100 m^{-1})*	2.8	1.1	ND	3.7	0.4	0.7	ND	4	0.8	0.6	ND	4
SRP (μg L^{-1})	7.4 (0.8)	7.3 (0.8)	7.1 (0.7)	7.2 (0.8)	4.6 (0.6)	5.8 (0.7)	5.0 (0.7)	5.9 (0.5)	4.2 (0.5)	5.9 (0.5)	3.9 (0.5)	6.1 (0.5)
NH_4^+-N (μg L^{-1})	8.6 (1)	7.2 (1)	8.8 (1)	7.5 (1)	4.9 (1)	6.8 (1)	5.3 (1)	15.9 (3)	6.2 (1)	15.9 (3)	5.5 (1)	14.4 (2)
Discharge (L s^{-1})	42.9 (7)	41.5 (7)	46.3 (6)	49.4 (5)	66.0 (4)	70.8 (7)	64.8 (9)	43.5 (13)	41.9 (12)	43.5 (13)	46.2 (10)	46.5 (10)
NO^-_3 (μg L^{-1})	177.0 (13)	159.5 (15)	174.8 (11)	164.6 (14)	148.5 (6)	171.6 (9)	149.7(6)	219.1 (25)	437.5 (83)	219.1 (25)	444.2 (81)	211.2 (25)
Sand (% cover)	28	32	28	35	11	19	14	23	18	36	28	28
Gravel (% cover)	27	18	23	19	29	33	25	23	23	25	29	26
Cobble (% cover)	5	4	7	4	22	16	22	19	13	10	9	9
Boulder (% cover)	<1	<1	0	<1	1	<1	3	2	5	1	1	1

*From Cordova *et al.*, 2007.

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accumulation, a ''grab'' (i.e. sub-sample) of organic matter was placed in a plastic bag, preserved in formalin, and processed for organic matter as described above and for macroinvertebrate production described below.

To convert debris accumulation volume into CBOM mass, we sacrificed and dismantled seven debris accumulations downstream of each 100-m treatment reach and repeated this each season to create a relationship between debris accumulation volume and CBOM mass. In the laboratory, we separated CBOM into leaves and small wood, which were dried and ashed as before. Predictive relationships were generated between debris accumulation volume and total CBOM, total leaf litter mass, and total small wood mass (excluding wood >10 cm in diameter) for each season. Total CBOM standing stock in debris accumulations for each stream reach was habitat-weighted by multiplying CBOM standing stock associated with ''new'' and ''old'' wood by their proportion covering the stream channel.

In-stream physical measurements

We surveyed benthic habitat using lateral transects spaced every 5–10 m (perpendicular to flow) in each stream in May and August 2003 before wood addition, and in May, July, and November 2004–06 (n = 11–21 transects per date in each reach) after wood addition. For each transect, we categorised substrate type every 10 cm across the wetted channel. Inorganic substrates were classified as boulders, gravel and cobble, or sand. Organic substrates were classified as moss, CBOM (<1 mm), FBOM, small wood (<10 cm) and large wood (>10 cm). We then calculated % cover for each substrate category at each transect; reach-level substrate % cover values represented means of each category across all transects on each date.

We also measured chlorophyll *a* on inorganic substrates (i.e. epilithon) by collecting 5 replicate 28.7-cm^2 cores from predominantly gravel habitat each month, except for January, in each reach from May 2004 to May 2006 (total of 22 intervals). We froze gravel samples for transport to the laboratory, and then extracted chlorophyll *a* using the fluorometric hot ethanol method (Sartory & Grobbelaar, 1984). We converted these values to an areal basis by weighting chlorophyll *a* concentration by the proportion of gravel in each reach.

We measured discharge monthly in each stream reach, with the exception of November–February when the streams had ice cover. We measured discharge and velocity from dilution of a conservative tracer during concurrent measurements of whole-stream nutrient uptake rates (Hoellein *et al.*, 2007). We recorded water temperature at the bottom of each stream reach hourly from May 2003 to May 2006 using HOBO® data loggers (Onset Computer Corporation, Bourne, MA, U.S.A.).

Macroinvertebrate secondary production – main channel

We sampled benthic macroinvertebrates within the main channel of the stream approximately monthly from May 2003 to April 2006 (31 intervals). Snow and ice cover prohibited sampling in December, January and February in the pre-treatment year (2003–04) and in January of the post-treatment years (2005–06). On each sampling date, in each stream reach, we randomly took five benthic samples using a 32-cm diameter Hess sampler (500-μm mesh in pretreatment year, 250-μm in post treatment years with no significant change in macroinvertebrate size-frequency distributions or densities, $P > 0.05$) driven down to 10 cm into the streambed when possible. Sampling methods were consistent among control and treatment reaches. This sampling excluded boulders and debris dams. Macroinvertebrates were preserved in 6–8% formalin and sorted from debris using a dissecting microscope (15× magnification). We identified macroinvertebrates to genus using keys in Merritt and Cummins (1996) and counted them, and measured their body length (nearest 1 mm). We assigned functional feeding groups to taxa using Merritt and Cummins (1996) and Wohl, Wallace & Meyer (1995). We estimated biomass using length–mass regressions in Benke *et al.* (1999). Secondary production was computed for common taxa using the size-frequency method with a correction for cohort production interval (CPI) obtained from size-frequency histograms (Benke, 1984; Benke & Huryn, 2006). For rare taxa, we estimated secondary production using published production to biomass (P/B) ratios (Krueger & Waters, 1983) or our own P/B ratios estimated from the specific study stream. We quantified chironomid production using the instantaneous growth method, where growth was estimated using temperature and size-specific growth rates in Walther *et al.* (2006).

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Macroinvertebrate secondary production – debris accumulations

Macroinvertebrates in debris accumulations were sampled in June 2004, November 2004 and April 2005 after wood addition. We do not have debris accumulation data before the wood was added. Macroinvertebrates per litter ''grab'' were expressed as number or biomass of macroinvertebrates per unit AFDM of grab. We averaged macroinvertebrate abundance and biomass among our three replicate grab samples in each stream reach on each sampling date, then multiplied mean abundance or biomass by the estimated amount of total organic matter in the reach, and finally divided by the total area of the accumulations. Secondary production was estimated for each sampling date, for each taxon, using P/B ratios calculated for the main channel taxa (Entrekin *et al.*, 2007). Density, biomass and production calculations were weighted by the areal proportion the accumulations represented within a reach relative to the main channel.

Data analysis

Physical differences among streams were elucidated with one-way analysis of variance (ANOVA). For all other variables measured with error, we used replicated before-after-control-intervention analysis-of-variance (BACI-ANOVA), blocked by stream, to detect treatment effects due to the experimental wood addition (Stewart-Oaten, Murdock & Parker, 1986; Underwood, 1992). Stream reaches to which we added large wood represented the intervention. When we observed significant interaction (BA*CI, $P < 0.05$), least squares means or Tukey's *post-hoc* tests were used to isolate differences in means. Spatial and temporal variation in secondary production was explored using simple linear regressions. We analysed all data using SAS® 9.1 software (SAS Institute Inc., Cary, NC, U.S.A.).

Because mean annual total secondary production is a single measure (i.e. no error), which incorporates secondary production in the main channel plus debris accumulations, we used a 3-factor analysis of variance, with stream, reach, and time since wood addition serving as factors. BACI-ANOVA could not be used because pretreatment data were not collected for debris accumulations. A significant interaction between stream reach and time since wood addition indicated a significant change in total secondary production between years after the treatment. Two-way repeated-measures ANOVA (RM-ANOVA) was used to evaluate differences in macroinvertebrate biomass between the main channel and debris accumulations in each stream if differences in biomass showed significant interaction between habitat type and month sampled. We transformed data when ANOVA assumptions were not met, using either a log or arcsine square-root transformation. Magnitude of change in macroinvertebrates is reported as the relative percent change between the control and treatment reaches before wood addition compared to the relative percent change between control and treatment reaches at 1 or 2 years after wood addition. We also calculated changes in P/B ratios, which are indicative of changes in either the average growth of taxa comprising the community or a change in biomass.

Results

Physical differences among study streams

Several physical characteristics differed consistently among the three streams over the course of the study regardless of wood addition. Canopy cover was dense in all streams, ranging from 73% in Walton Creek to 82% in Shane Creek. In all streams, canopy cover was greater in spring and summer (mean, 88 ± 1%) relative to winter (mean, 73 ± 3%) (ANOVA $F_{9,96} = 5.78$, $P < 0.001$). Compared to Walton Creek and Shane Creek, State Creek ~100 more cumulative degree days during the winter and early spring. State Creek also had significantly less sand ($F_{5,48} = 15.71$, $P < 0.001$), more cobble and gravel ($F_{5,48} = 3.87$, $P < 0.001$), a lower volume of large wood, and less annual variability in discharge than Shane Creek and Walton Creek (Table 1).

Macroinvertebrate secondary production – main channel

Two years after wood addition, we measured no significant change (average of all streams) in overall macroinvertebrate number of taxa, density, or biomass (Table 2, BACI-ANOVA, $P > 0.05$) in the treatment reaches compared to the control reaches because the macroinvertebrate communities responded differently

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Table 2 Average (±SE) annual taxa richness (no. m^{-2}), density (m^{-2}), and biomass (mg AFDM m^{-2}) in each stream reach 1 year before wood addition (Y0) and one (Y1) and two (Y2) years after wood addition

	Y0			Y1			Y2		
	C	T	T − C	C	T	T − C	C	T	T − C
No. of taxa									
Shane Creek	31.0 (1.5)	31.1 (2)	0.1	29.9 (2)	29.5 (2)	−0.3	32.3 (1)	32.6 (1)	<1
State Creek	28.3 (1.3)	27.5 (2)	−0.7	27.7 (1)	24.8 (1)	−2.9	31.1 (0.8)	31.0 (1)	<1
Walton Creek	**30.2 (1)**	**32.1 (1)**	1.8	**35.3 (1)**	**29.9 (1)**	−5.4	**38.7 (1)**	**35 (1)**	−3
Mean	29.8 (0.7)	30.2 (1)	0.4	31.0 (34)	1.1 (0.9)	−29.9	28.1 (32)	1.1 (0.9)	−27
Density									
Shane Creek	1060.6 (118.7)	1127.1 (175)	66.4	2253.5 (331)	2203.6 (225)	−49.9	2552.1 (390)	2463.0 (240)	−89
State Creek	1542.2 (237.1)	1158.3 (130)	−384	2533.2 (300)	1691.1 (168)	−842.1	2850.3 (381)	2901.0 (253)	50
Walton Creek	**1117.0 (106)**	**1343.8 (146)**	**227**	2072.4 (204)	1817.2 (228)	−255.1	**4136.3 (405)**	**3030.0 (279)**	**−1106**
Mean	1241.1 (97.6)	1212.5 (88)	−29	2284.0 (161)	1905.2 (121)	−378.7	3148.4 (231)	2796.6 (149)	−351
Biomass									
Shane Creek	248.4 (35.3)	225.1 (34)	−23	394.6 (74)	217.5 (26)	−177.0	263.3 (75)	171.0 (27)	−92
State Creek	**501.9 (89.8)**	**332.1 (51)**	**−170**	373.3 (39)	342.7 (39)	−30.5	**278.9 (36)**	**405.2 (47)**	**126**
Walton Creek	372.1 (62.5)	239.0 (23)	−133	376.5 (44)	293.7 (34)	−82.8	345.8 (44)	181.5 (25)	−164
Mean	367.4 (37.7)	265.7 (22)	−102	381.2 (30)	283.2 (19)	−97.9	295.3 (31)	253.9 (21)	−41

Bolded numbers represent a significant difference in the treatment (T) values compared to the control (C) before and after wood addition (BACI factorial ANOVA). T − C is the difference where a positive indicates a higher value in the treatment reach and a negative indicates a higher value in the control reach.

to wood addition in each stream. In Walton Creek, the number of macroinvertebrate taxa declined by 5 genera (4%) (Table 2, Y1 $F_{3,36} = 7.07$, $P = 0.011$; Y2 $F_{3,36} = 4.60$; $P = 0.03$) and density declined by 1333 m^{-2} (12%) (Table 2, Y1 $F_{3,171} = 2.71$, $P = 0.10$; Y2 $F_{3,180} = 3.90$, $P = 0.04$) 2 years after wood addition. In contrast, number of taxa did not change over time in Shane Creek or State Creek and was similar between reaches before and after wood additions. In State Creek, biomass increased by 296 mg AFDM m^{-2} (5%) 2 years after wood addition (Y2 $F_{3,189} = 5.36$, $P = 0.02$) compared to the control.

When all three streams were considered together, macroinvertebrate secondary production was not significantly altered by the wood addition (Fig. 2a), likely because individual streams responded

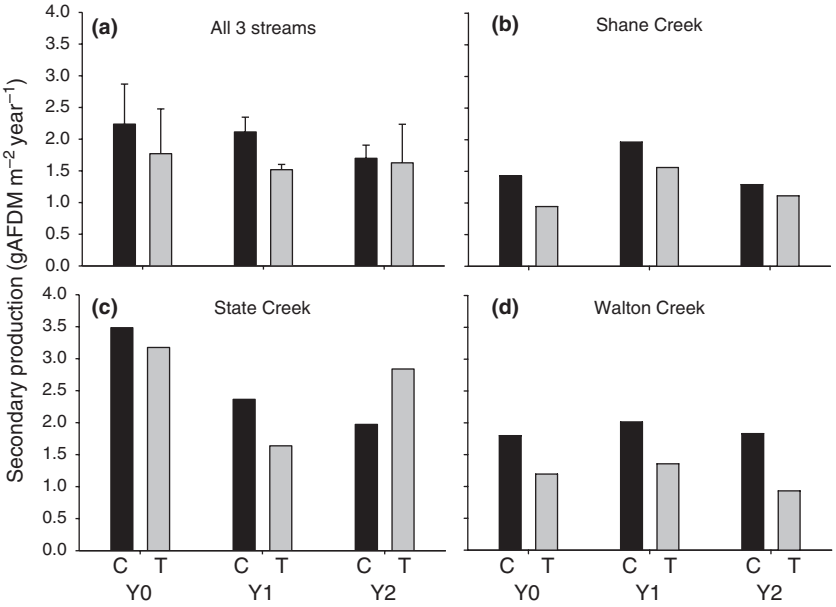

Fig. 2 Annual mean secondary production in the control (C) and treatment (T) reaches before (Y0) and 1 year (Y1) and 2 years (Y2) after wood addition. (a) Total mean production in all streams; (b) production in Shane Creek; (c) production in State Creek and (d) production in Walton Creek.

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differently to the manipulation. Relative to the control, secondary production in the treatment reaches increased by 388 mg AFDM m^{-2} (12%) in Shane Creek (Fig. 2b) and 1100 mg AFDM m^{-2} (22%) in State Creek (Fig. 1c) in year 2 after wood addition. In contrast, secondary production declined by 300 mg AFDM m^{-2} (12%) in the treatment reach of Walton Creek compared to the control over the same period (Fig. 2d). We measured only small, insignificant changes in P/B ratios following the addition of wood, although changes in P/Bs differed by streams (Table 3). Ratios ranged from 4.2 in Walton Creek, indicating more slow-turnover taxa, to 9.6 in State Creek.

We then examined the relative contribution of different functional feeding groups to total production following wood addition. Shredder production changed very little over time because of different responses in individual streams (Table 4). Across streams, Capniidae (Plecoptera) significantly increased in the treatment reaches 1 year after wood addition ($F_{5,6}$ = 7.84, P = 0.03) and two leaf-shredding stoneflies, *Amphinemura* sp. and *Allocapnia* spp., tended to increase 2 years after wood addition ($F_{5,6}$ = 4.60, P = 0.07). In Shane Creek and Walton Creek, shredder production was variable among years, but did not change in response to the addition of wood.

Averaged among streams, secondary production of scrapers significantly declined at 1 year after wood addition (Table 4), but increased to higher than pre-wood addition values in year 2 (Table 4, Y1 $F_{5,6}$ = 7.82, P = 0.03; Y2 $F_{5,6}$ = 0.01, P = 0.91). The overall positive scraper response was driven by State Creek, where scraper production in the treatment reach decreased by *c.* 400 mg AFDM m^{-2} $year^{-1}$ (or 24%) relative to the control reach 1 year after wood addition, but increased by >600 mg AFDM m^{-2} $year^{-1}$ (or 22%) in the treatment relative to the control reach in year two. Declines in *Glossosoma* sp. drove the decrease in scraper production in year 1 ($F_{5,6}$ = 21.78, P = 0.003), whereas *Baetis* drove the increase in State Creek in year 2 (Appendix S1).

Overall gatherer production did not change significantly in either year after wood addition, but tended to decline in the treatment reaches of State Creek and Walton Creek in year 2 after wood addition by nearly 300 mg AFDM m^{-2} day^{-1} (Table 4). Overall filterer production in the treatment reaches also did not change significantly relative to the control reaches, but production increased in Shane Creek by *c.* 300 mg AFDM m^{-2} $year^{-1}$ (or 8%) in year 2 relative to the control reach. In State Creek, filterer production in the treatment reach increased by nearly 600 mg AFDM m^{-2} day^{-1} (or 17%) in year 2 relative to the control reaches. Increases in filterer production in the treatment reach were due, in part, to a significant increase in the production of the blackfly genera *Simulium* and *Prosimulium* spp. 2 years after wood addition ($F_{5,6}$ = 6.48, P = 0.03). Filterer production did not change in Walton Creek (Table 4).

Predator production was high compared to other functional groups and represented by 20 taxa. Production of two predators, the stonefly *Isoperla* spp. ($F_{5,6}$ = 5.04, P = 0.06) and the cranefly *Hexatoma* spp. ($F_{5,6}$ = 5.36, P = 0.05), increased 1 year after wood addition. The caddisfly *Rhyacophila* spp. ($F_{5,6}$ = 9.56, P = 0.01) and the cranefly *Dicranota* sp. ($F_{5,6}$ = 17.33, P = 0.003) increased 2 years after wood addition. Predator production changed little in the Shane Creek treatment reach relative to the control. In State Creek, predator production declined by *c.* 200 mg AFDM m^{-2} $year^{-1}$ in the treatment reach relative to the control reach 1 year after the wood addition and then

Table 3 Annual production to biomass ratios (P/B) in the main channel of each stream

	Y0			Y1			Y2		
	C	T	T − C	C	T	T − C	C	T	T − C
Main channel									
Shane Creek	5.8	4.2	−1.6	5.0	7.2	2.2	4.9	6.5	1.6
State Creek	6.9	9.6	2.6	6.3	4.8	−1.6	7.1	7.0	−0.1
Walton Creek	4.8	5.0	0.2	5.4	4.6	−0.7	5.3	5.1	−0.2
Mean	6.1	6.7	0.6	5.5	5.4	−0.2	5.7	6.4	0.7

T (treatment) − C (control) is the magnitude of difference where a positive value indicates a higher value in the treatment reach and a negative indicates a higher value in the control reach.

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Table 4 Average (±SE) functional feeding group secondary production (mg AFDM m^{-2} y^{-1}) in each stream reach before (Y0) and one (Y1) and two (Y2) years after wood addition

	Y0			Y1			Y2		
Feeding group	C	T	T − C	C	T	T − C	C	T	T − C
Shredders									
Shane Creek	92.2	59.1	−33	80.9	57.5	−23	46.8	59.1	12
State Creek	45.8	112.2	66	124.7	60.5	−64	133.7	202.1	68
Walton Creek	89.3	42.3	−47	124.2	92.6	−32	63.6	39.0	−25
Mean	75.7 (15)	71.2 (21)	−5	109.9 (14)	70.2 (11)	−39	81.3 (26)	100.1 (51)	18
Scrapers									
Shane Creek	181.0	220.6	40	324.7	207.9	−117	287.0	190.5	−96
State Creek	1010.9	1253.4	243	817.3	604.5	−213	899.1	1752.0	853
Walton Creek	378.3	559.3	181	548.9	270.1	−279	410.9	218.4	−193
Mean	**523.4 (250)**	**677.7 (142)**	**154**	**563.6 (142)**	**360.8 (123)**	**−203**	**532.3 (186)**	**720.2 (515)**	**187**
Collector-filterers									
Shane Creek	364.2	202.9	−161	613.1	736.8	124	211.8	362.6	151
State Creek	1570.5	1081.9	−489	404.1	399.4	−5	192.6	271.9	79
Walton Creek	475.8	197.6	−278	467.2	192.8	−274	357.0	147.0	−210
Mean	803.5 (384)	494.1 (61)	−309	443.0 (158)	253.8 (51)	−189	253.8 (51)	260.5 (62)	7
Collector-gatherers									
Shane Creek	153.4	134.1	−19	163.1	136.2	−27	198.1	171.2	−27
State Creek	131.3	146.9	16	253.7	243.8	−10	370.0	157.3	−213
Walton Creek	63.0	48.8	−14	261.4	177.6	−84	534.5	234.1	−300
Mean	115.9 (27)	109.9 (30)	−6	226.1 (31)	185.9 (31)	−402	367.5 (97)	187.5 (23)	−180
Predators									
Shane Creek	637.8	326.4	−311	745.6	410.7	−335	542.7	295.6	−247
State Creek	725.5	580.4	−145	750.8	323.8	−427	364.6	441.0	76
Walton Creek	791.8	347.4	−444	608.8	611.4	3	439.2	271.8	−167
Mean	718.4 (44)	418.0 (81)	−300	701.7 (46)	448.6 (85)	−253	448.8 (51)	336.1 (52)	−113

Bolded numbers represent a significant difference in the treatment (T) values compared to the control (C) before and after wood addition (BACI factorial ANOVA). T − C is the magnitude of difference where a positive indicates a higher value in the treatment reach and a negative indicates a higher value in the control reach.

increased by *c.* 250 mg AFDM m^{-2} $year^{-1}$ 2 years after wood addition. Predator production in Walton Creek increased by *c.* 450 mg AFDM m^{-2} $year^{-1}$ after 1 year, resulting in very little difference between the control and treatment reaches, but returned to pretreatment levels in year 2 (Table 4).

In State Creek, production of multivoltine macroinvertebrates was lower in the treatment reach than the control before wood addition, increased by 750 mg AFDM m^{-2} $year^{-1}$ 1 year after wood addition, and exceeded pre-wood production values relative to the control reach by 500 mg AFDM m^{-2} $year^{-1}$ 2 years after wood addition (Fig. 3a). Production of univoltine taxa was greater in the treatment reach by 300 mg AFDM m^{-2} $year^{-1}$ than the control before wood addition, declined by 650 mg AFDM m^{-2} $year^{-1}$1 year after wood addition, and increased back to pre-wood conditions relative to the control (Fig. 3b).

Relationship to habitat variables

Changes in substrate percent cover were not significant when data were pooled annually and analysed using BACI-ANOVA (Table 1, $P > 0.05$). However, we did observe changes in substrate distributions due to wood addition, particularly in Shane Creek and State Creek, where sand deposited around the newly added logs. We also found several strong relationships between substrate percent cover and macroinvertebrate secondary production. Total secondary production of the macroinvertebrate community was negatively related to benthic sand coverage (Fig. 4a, $r^2 = 0.59$, $P < 0.01$). In addition, scraper production, which declined at 1 year and increased at 2 years after wood addition, was positively related to % cobble and gravel (Fig. 4b, $r^2 = 0.54$, $P < 0.01$) and negatively related to % sand cover ($r^2 = 0.46$, $P < 0.01$, data not shown). The increase in total secondary production in

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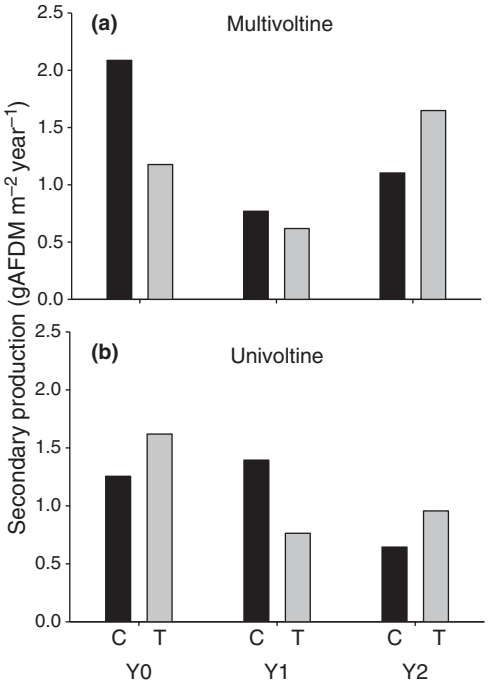

Fig. 3 Annual mean secondary production of (a) multivoline and (b) univoltine macroinvertebrate taxa in control (c) and treatment (T) reaches before (Y0) and 1 year (Y1) and 2 years (Y2) after wood addition in State Creek.

State Creek was driven, partly, by an increase in exposed cobble and gravel (Fig. 4c). Mean annual macroinvertebrate abundance increased in the treatment reach of Walton Creek, but not as much as in the control reach (Fig. 4d). The increase in abundance was related to an increase in fine benthic organic matter (FBOM) standing stocks, which declined in the treatment reach, but not the control reach after wood addition (Fig. 4d).

Macroinvertebrate secondary production – debris accumulations

Accumulations of organic matter represented a small percentage of the total available benthic habitat on an areal basis (<2.2%), but CBOM standing stocks in the main channel were low in our study streams compared to levels in other streams (Entrekin *et al.*, 2007; Hoellein *et al.*, 2007). Therefore, debris accumulations may be particularly important in these streams and we began sampling macroinvertebrates in the accumulations when the wood was added (i.e. May 2004). Leaves and small wood in debris accumulations represented a substantial amount of the total organic matter within the streams (Entrekin *et al.*, 2008) and the macroinvertebrate secondary production within the debris accumulations was also high compared to main channel values (Fig. 5). Annual macroinvertebrate secondary production in the debris accumulations exceeded secondary production in the main channel, when the values from all three streams were averaged, by as much as 5300 mg AFDM m^{-2} year^{-1} (44%, Fig. 2). Similar to main channel secondary

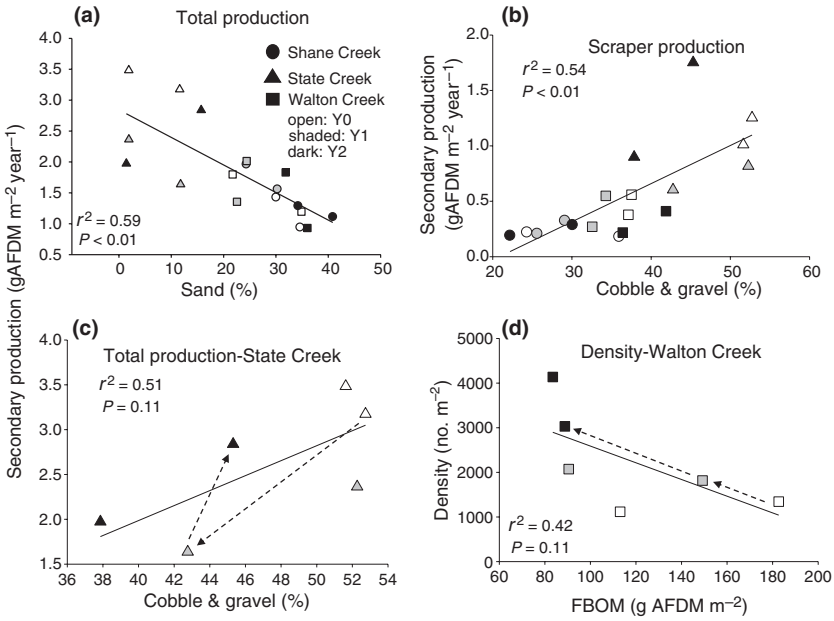

Fig. 4 Simple linear regressions relating significant changes in macroinvertebrate metrics, due to wood additions, to changes in substrate composition and fine organic matter standing stocks. (a) Mean annual changes in production related to changes in percent cover of sand, (b) mean annual scraper production related to changes in gravel and cobble percent cover, (c) total annual production in State Creek only related to changes in cobble and gravel percent cover and (d) total annual production in Walton Creek related to changes in standing stock of fine benthic organic matter (FBOM). Arrows indicate changes that occurred in the treatment reach following wood addition.

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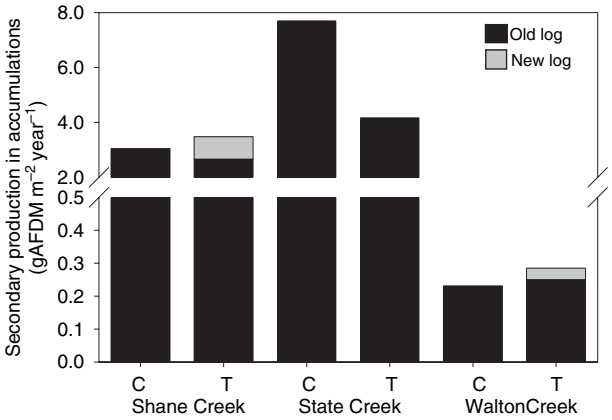

Fig. 5 Macroinvertebrate production in debris accumulations separated by production associated (assc.) with old wood and newly added logs 1 year after (Y1) wood addition in Shane Creek, State Creek, and Walton Creek.

production, debris accumulation production responses varied among streams (Fig. 5). In both Shane Creek and State Creek, macroinvertebrate production in debris accumulations was greater than in the main channel at 1 year. In Walton Creek, secondary production in debris accumulations was lower than in the main channel after year 1 (Fig. 5).

Macroinvertebrate biomass differed between benthic and debris accumulations (Fig. 6; $F_{2,3} = 1.16$, $P = 0.02$). Differences in production among macroinvertebrates in benthic and debris accumulations were due, in part, to differences in monthly mean biomass and taxonomic composition (Figs 5 & 6). Mean biomass was variable among months in both habitat types and streams. In Shane Creek, biomass was greater in the main channel in June (Fig. 6a; $F_{1,2} = 28.18$, $P = 0.03$) and greater in debris accumulations in November ($F_{1,2} = 25.07$, $P = 0.03$). In State Creek, biomass was greater in the main channel in June (Fig. 6b; $F_{1,2} = 34.72$, $P = 0.02$) and April ($F_{1,2} = 78.32$, $P = 0.01$), but greater in debris accumulations in November ($F_{1,2} = 29.04$, $P = 0.03$). In Walton Creek, biomass was always greater in the main channel than debris accumulations regardless of month (Fig. 6c; $F_{5,6} = 22.59$, $P < 0.01$).

The composition of functional feeding groups differed significantly between the main channel and debris accumulations (Fig. 7a,b). Scraper production represented a larger percentage (mean, 25 ± 4%) of the total secondary production in the main channel than in debris accumulations (mean, 5 ± 2%, $P < 0.0001$). In contrast, shredder production represented a larger percentage of total production in debris accumulations (mean, 23 ± 9%) than in the main channel (mean, 5 ± 0.5%, $P = 0.03$). Neither collector nor predator production differed between the main channel and debris accumulations in percentage of total production (Fig. 7a,b).

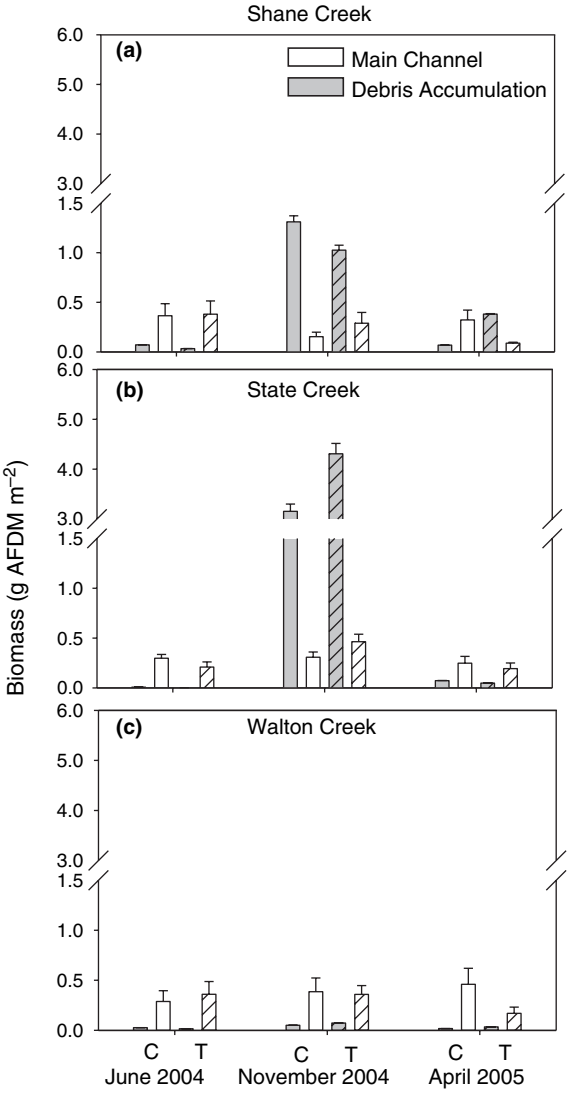

Fig. 6 Comparing mean macroinvertebrate biomass in benthic and debris accumulations for the months debris accumulations were sampled in control (C) and treatment (T) reaches in (a) Shane Creek, (b) State Creek and (c) Walton Creek.

Discussion

The experimental addition of wood to three headwater streams produced variable responses in

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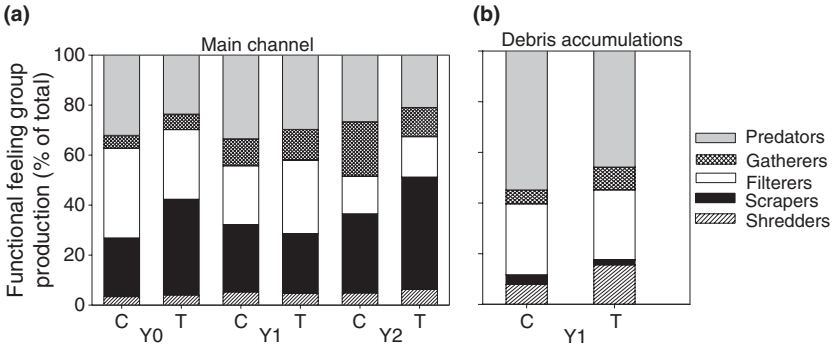

Fig. 7 Secondary production of each functional feeding group as percent of total in the control and treatment reaches in each of the three study streams in the (a) main channel and (b) debris accumulations 1 year after wood addition.

macroinvertebrate community structure and production in two important habitats; the main channel and debris accumulations (e.g. Roeding & Smock, 1989; Smock *et al.*, 1989). Macroinvertebrate responses in the main channel appeared to be mediated mostly by substrate composition, while the retention of leaf litter strongly influenced production associated with accumulations. Macroinvertebrate production was very low before wood addition and declined in the first year following wood addition. The decline 1 year after large wood addition seemed to have acted as more of a disturbance to univoltine biota in all streams (e.g. Muotka *et al.*, 2002), but by year 2 macroinvertebrates recovered and responded positively to wood addition in State Creek.

Main channel secondary production

Secondary production of macroinvertebrates has been shown to be mediated by local geomorphology and temperature (Huryn & Wallace, 1987; Bourassa & Morin, 1995; Wohl *et al.*, 1995). Stable substrata are critical in structuring macroinvertebrate communities at a local scale (Pringle *et al.*, 1988) and secondary production often increases with the availability of stable substrate by providing biota areas to forage and grow (e.g. Benke *et al.*, 1992; Benke & Wallace, 2003). Macroinvertebrate responses to wood addition in the main channel were mediated by the interaction among wood, streamflow, water temperature and substrate composition. Changes in substrate distribution associated with wood additions resulted in an overall increase in scraper secondary production across streams, a decline in the abundance and diversity of macroinvertebrates in Walton Creek, and an increase in total macroinvertebrate production in State Creek.

Wood affected sediment distributions by altering local water velocities, which appeared to result in a decline of most taxa relative to the control reaches during the first year (see also Muotka *et al.*, 2002). Over the longer term, substrate sorting and increased gravel and cobble exposure may occur, which could increase habitat for macroinvertebrates and periphyton growth (Lamberti & Resh, 1979; Reice, 1980). In addition, colonisation of macroinvertebrates on the wood itself should be an important stable habitat, although significant colonisation (Benke & Wallace, 2003) was not apparent during this study (S. A. Entrekin, pers. obs.). Although these streams have substantial canopy, primary production may be important to the food web for much of the year (Entrekin *et al.*, 2007). For example, increases in main-channel secondary production in State Creek were related to concurrent increases in gravel and cobble exposure and a reduction in fine-grained sediments. Because these streams flow through fine sediment geology, continued exposure of coarse sediments and a reduction in fine sediments through substrate sorting may eventually lead to increases in main-channel production in the other two streams. However, more data are needed to determine if added structure will result in increased macroinvertebrate production, or if other factors, such as water temperatures and low nutrient conditions are also limiting production.

We propose several reasons that the main-channel macroinvertebrate community in the three streams responded differently to wood additions. First, changes in secondary production in the main channel were related to changes in particle size distribution (see also Kail, 2002). Most sediments were highly mobile because of their small size. Substrate composition in State Creek, however, was larger and

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likely more resistant to streamflow change, which may have enabled us to detect subtle changes in the macroinvertebrates resulting from wood additions. State Creek also had less variable discharge and warmer winter water temperatures due to groundwater inputs (see also Krueger & Waters, 1983; Eggert & Burton, 1994). Although the absolute increase in winter water temperature was slight in State Creek (<0.5 °C), it translated into 100 additional degree days over the growing season that kept a greater percentage of the stream ice-free, promoting algal growth (Entrekin *et al.*, 2008). Differences in surrounding land use may have also been a factor. Shane Creek and State Creek are in 40-year-old forests, whereas Walton Creek is in a 90-year-old forest. Therefore, the banks of Walton Creek may be more stable and more resistant to changes by the added wood or other stable substrata.

Debris accumulation secondary production

Increased retention of organic matter is often a goal of wood additions to forested streams (e.g. Lemly & Hilderbrand, 2000; Lepori *et al.*, 2005; Rosi-Marshall, Moerke & Lamberti, 2006) because organic matter fuels heterotrophic production. Prior to manipulation, our study streams had low amounts of wood (Cordova *et al.*, 2007), low quantities of benthic organic matter and low macroinvertebrate secondary production (Entrekin *et al.*, 2007) compared to other temperate forested streams (reviewed by Huryn & Wallace, 2000); therefore, we predicted that wood addition would increase organic matter retention and lead to direct bottom-up effects on the macroinvertebrate community (see also Richardson, 1991).

Prior to wood addition, debris accumulations around large wood were a very small portion of the entire stream channel, but the relative importance of such accumulations might be substantial, especially in autumn, for several reasons. First, shredding macroinvertebrates were sparse in the main channel but organic matter was processed at a high rate (Entrekin *et al.*, 2008). Second, we observed aggregations of shredders in debris accumulations (see also Kobayashi & Kagaya, 2005). Third, because secondary production in the main channel was low and production in debris accumulations contributed substantially to total secondary production in autumn and sometimes in spring.

Debris accumulations as resource islands

Debris accumulations supported a more productive community of shredders and predators than did the main channel, probably due to the differences in available resources. Gatherer (e.g. chironomid midges and *Ephemerella*) and filterer (e.g. blackflies and net-spinning caddisflies) production tended to represent the same proportion of total production in both habitats due to their generalist feeding ability (Cummins & Klug, 1979; Wallace & Webster, 1996). High rates of macroinvertebrate production within the accumulations relative to the main channel suggest that they are hotspots of carbon processing in autumn and winter. Debris accumulations buffer the entire macroinvertebrate community from disturbance and act as important sources of habitat and food (e.g. Palmer *et al.*, 1996). As the new logs accumulate more organic debris, additional allochthonous carbon should move up the food web to fish or leave the streams via invertebrate emergence. Debris accumulations were clearly an important component of the food web in these streams and were probably much more numerous when the catchments contained mature forest and beaver were common (see also Bilby & Likens, 1980; Likens & Bilby, 1982; Smock *et al.*, 1989).

Several previous large-scale experimental manipulations have assessed changes in macroinvertebrate production through bottom-up effects (Wallace *et al.*, 1999; Wilcox *et al.*, 2005; Cross *et al.*, 2006). For example, macroinvertebrate secondary production declined by more than 50% after an experimental exclusion reduced leaf litter and small wood standing stocks by around 95% (Wallace *et al.*, 1999). In contrast, macroinvertebrate detritivore and predator production increased by more than 200% after 3 years of nutrient enrichment with inorganic N and P (Cross *et al.*, 2006). Macroinvertebrate communities in one of our study streams displayed an increase in short-lived taxa, such as *Baetis* mayflies, that responded quickly (<1 year) to changes in habitat and resources in the main-channel. Reductions in secondary production, however, tended to occur among long-lived taxa such as caddisflies and stoneflies. Wood additions seemed to act as a disturbance to univoltine taxa, such as *Glossosoma*, in the first year but not to mulivoltine taxa, such as *Baetis* and Chironomids. Univoline production increased years

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after wood addition, indicating recovery. The variable response by taxa with different life histories emphasises the importance of evaluating restoration success in many ways.

In State Creek, total secondary production increased by 22%. The increase in secondary production could have been as much as 45% in both the main channel and in production around newly added logs if P/B in accumulations was assumed to be the same between years 1 and 2. Macroinvertebrates responded to both geomorphic changes in the stream channel and appeared to track leaf litter accumulations (e.g. Hall, Likens & Malcom, 2001; Negishi & Richardson, 2003). For the newly added logs to have a reach-scale effect on macroinvertebrate secondary production in the other streams, changes in geomorphology and organic matter must be greater. Leaf litter inputs and organic matter standing stocks in these streams were comparable to streams in other temperate deciduous forests (e.g. Hall *et al.*, 2001; Chadwick & Huryn, 2005). However, the relationship between macroinvertebrate production and leaf litter was more variable and production was very low, probably due to colder winters, less stable sediments and more variable streamflows (Fig. 8; Entrekin *et al.*, 2007). Still, as small and large wood becomes retained on the added logs to form debris accumulations, leaf litter retention, organic matter processing and macroinvertebrate production are likely also to increase.

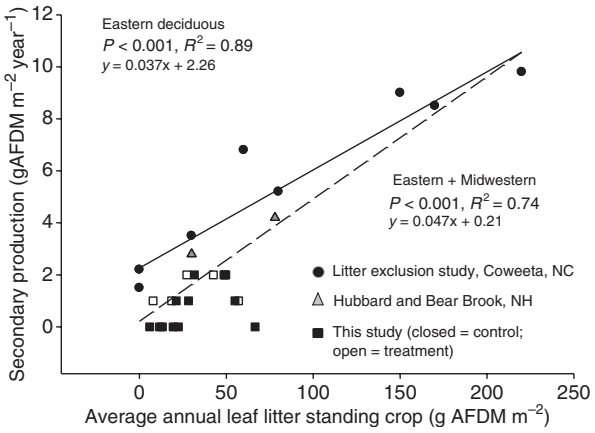

Fig. 8 Relationship between macroinvertebrate secondary production rates and annual leaf litter standing crop in the main channels of perennial, forested, headwater streams (data from Wallace *et al.*, 1999; Hall *et al.*, 2001; and Chadwick & Huryn, 2005). Our estimates of total secondary production (main channel plus debris accumulations) are plotted (black rectangles = control, open rectangles = treatment), but data were not included in the top regression calculations (solid line). We conducted another regression with all data to illustrate the difference in the relationship between leaf litter and production in our study streams compared to the other streams.

It may take many years for organic matter accumulation to have an overall positive effect on the macroinvertebrate communities, as other environmental variables also influence secondary production in these streams. In small temperate forested streams, water temperature, extent of ice cover, flooding and primary production are also important variables that will influence total secondary production and community structure (e.g. Iversen, 1988; Berg & Hellenthal, 1992; Mccutchan & Lewis, 2002). Future research should examine the interaction among wood additions and ice cover extent, flooding and primary production.

Comparison to other stream restorations

Large wood increases habitat heterogeneity, structural complexity and niche space, which should lead to increases in species diversity and invertebrate density, and possibly shifts in community composition (e.g. Likens & Bilby, 1982; Johnson *et al.*, 2003). However, a review of 11 wood or boulder additions that had the stated goal of increasing habitat heterogeneity or organic matter retention and included invertebrate responses did not support this hypothesis (Appendix S2). Unfortunately, detailed invertebrate data were lacking for most addition studies; total densities were most commonly recorded, followed by species diversity metrics, and biomass expressed as functional groups or individual taxa more rarely. Macroinvertebrate density responded positively in only six of the 11 studies. Species richness was reported in only six studies, and only one (Gerhard & Reich, 2000) showed a positive increase in taxon richness. Results from our study of three streams also revealed considerable variation in macroinvertebrate responses, with density and taxon richness declining in one stream and not changing in the other two streams. Functional composition was measured in only two other studies (Lemly & Hilderbrand, 2000; Moerke & Lamberti, 2004) and in both was related to changes in substrate composition. In other studies, the alteration of water velocities and scour also led to increased pool formation or deposition of sediment and organic

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matter (Wallace *et al.*, 1995; Gerhard & Reich, 2000). Overall, inconsistencies were found between the hypothesis that adding large wood or boulders will increase diversity and enhance ecosystem function and the published field data that actually show such changes.

Effects of wood addition on secondary production

Secondary production has been measured following the addition of large wood in only one other stream to our knowledge. Wallace *et al.* (1995) reported an increase in detritivore production after wood was added to a second order Appalachian stream. In contrast, we found an increase in scraper production and little change in detritivore production. Several factors may account for these differences. Although the streams in both studies were similar in size, log placement was different. In the Appalachian stream, logs were placed perpendicularly to streamflow and spanned the entire channel, acting as dams that increased sediment and organic matter accumulation on the upstream side and accelerating shear stress on the downstream side. These logs generally changed riffle habitat to depositional habitat and functional group structure changed accordingly. Total secondary production increased in the Appalachian stream because longer-lived taxa were replaced by chironomid midges with high turnover rates. In our study, we placed logs haphazardly into the streams, generally with part of the log resting on the stream bank, and never spanning the entire channel. Therefore, in changes in geomorphology were more subtle, and extensive depositional habitat, which would enhance detritivory, was not created and little change occurred in turnover rates.

Large wood additions can increase macroinvertebrate secondary production in two ways: by increasing suitable habitat or retaining food resources. We found positive changes in both habitat and resources in one stream, but little effect and even negative effects on habitat, resources and macroinvertebrates in the other two streams. We measured changes in production for 2 years following wood addition, but changes in habitat and increases in organic matter storage may require much longer to unfold. If these processes occur, they may lead to increased secondary production in such poorly retentive and habitat-poor streams. Therefore, many years may be required to see significant and sustainable positive effects on the aquatic food web in response to wood addition. A better practice may be to limit the removal of wood from streams and from the surrounding catchment to maintain ecosystem function. Our results illustrate the importance of sustainable catchment management and the complexity of restoring ecosystem function to previously altered streams.

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Supporting Information

Additional Supporting Information may be found in the online version of this article:

Appendix S1. Macroinvertebrate production (mg AFDM m^{-2} $year^{-1}$) of each taxon in each study stream.

Appendix S2. Summary of stream and river restorations that added large wood or boulders.

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